LEARNING SESSION S7

AI, Generative AI & Agentic AI (for non-tech learners)

How AI, LLMs, RAG and AI agents work, what they are good and bad at, and how to use them safely at work

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

Interactive version (every tab clickable):
https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw

Contents

The big picture
How LLMs work
RAG & fine-tuning
Agentic AI
Using AI at work
Interview Q&A
Resources

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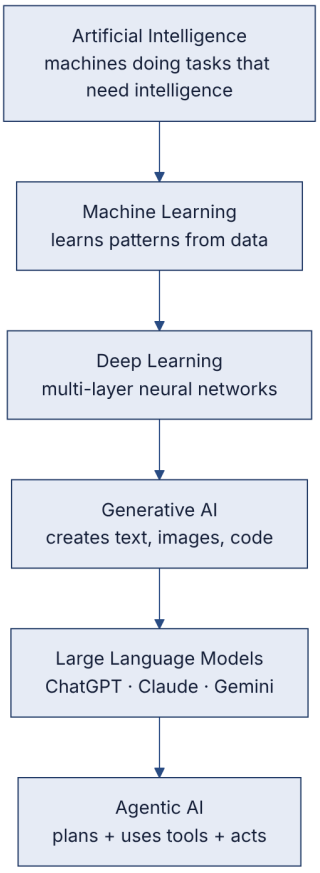
🧠 AI, Generative AI & Agentic AI (for non-tech learners)

How AI, LLMs, RAG and AI agents work, what they are good and bad at, and how to use them safely at work

Learning session · no technical background needed · complements the Generative AI course module

Open this module online: **https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw#s7**

The big picture



Each layer is a subset of the one above: agentic AI is the newest layer, built on LLMs.

Term	Plain meaning	Example
Artificial Intelligence	Machines doing tasks that normally need human intelligence	Chess engines, voice assistants
Machine Learning	Learns patterns from data instead of hand-written rules	Credit scoring, fraud alerts
Deep Learning	ML with large neural networks	Face unlock, speech-to-text

Term	Plain meaning	Example
Generative AI	Creates new content: text, images, audio, video, code	ChatGPT, Claude, Midjourney
LLM	Large language model trained on huge text to predict the next word	GPT, Claude, Gemini, Llama, Grok
Agentic AI	AI that plans steps and uses tools to complete a goal, checking its own progress	"Pull last week's sales, build the chart, draft the email"

A short timeline

1950s–1990s	Rule-based AI, early neural networks, expert systems
2010s	Deep learning breakthroughs in images and speech
2017	The Transformer architecture ("Attention Is All You Need")
Nov 2022	ChatGPT launch brings generative AI to everyone
2023–2024	Multimodal models (text + images + voice), copilots in Office and coding tools, RAG in enterprises
2025–2026	Reasoning models, AI agents that browse, code and use apps; agent standards such as the Model Context Protocol (MCP); AI inside Excel, Power BI and Tableau

How LLMs work



An LLM writes one token at a time, each time predicting the most suitable next piece of text.

An analogy

Imagine someone who has read most of the internet and can finish any sentence plausibly. They are brilliant at language and patterns, but they don't "look up" facts unless given a source, and they can sound confident while wrong. That's an LLM.

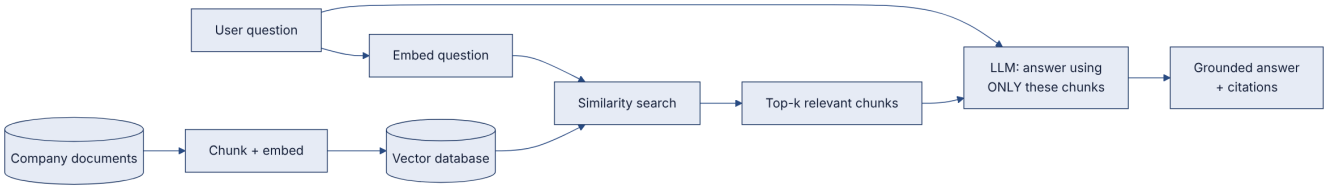
Token	A word piece; pricing and limits are in tokens (~¾ of a word each)
Context window	How much text the model can consider at once (a few pages to entire books)
Training	Pre-training on huge text → instruction tuning → human-feedback tuning for helpfulness and safety
Temperature	Creativity dial: low for facts and code, higher for ideas

Hallucination	A fluent but false answer; always verify numbers, names and citations
Knowledge cut-off	The model knows little after its training date unless it can search
Multimodal	Reads images, PDFs, spreadsheets; speaks and listens
Reasoning models	Spend extra "thinking" steps on hard problems (maths, code, planning)

Prompting techniques (role, format, chain of thought, ask-before-answer, reverse prompting) are covered in **Generative AI → Prompting**.

RAG & fine-tuning

Retrieval-Augmented Generation (RAG)



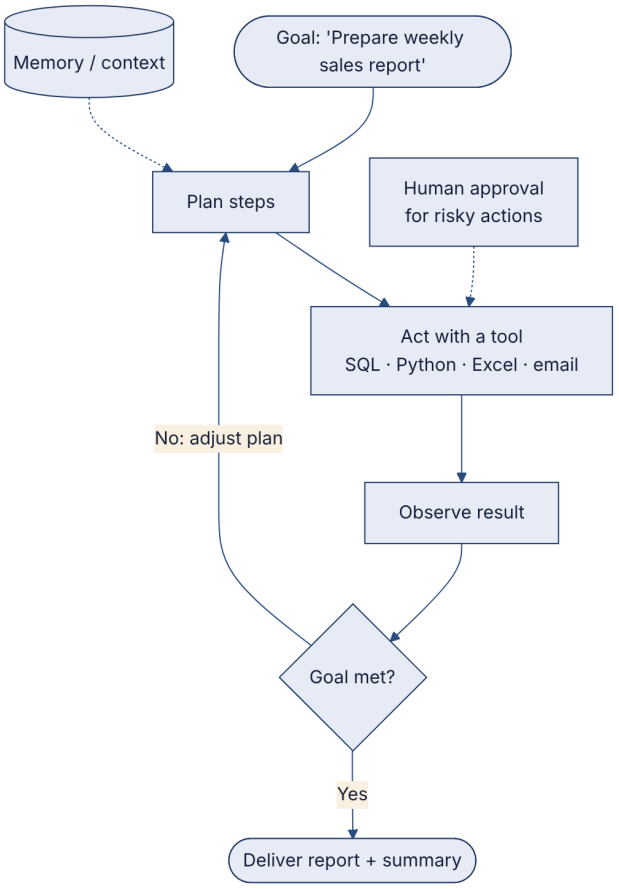
RAG lets an LLM answer from your own documents: it searches first, then writes the answer using only what it found.

- **Embeddings** turn text into number vectors so that similar meanings are close together (cosine similarity, see **Building Blocks ML · Class 7**).
- A **vector database** stores those vectors for fast similarity search (Pinecone, Chroma, pgvector, and now SQL Server 2025's VECTOR type).
- Example: an HR bot that answers "How many casual leaves do I get?" from the company's policy PDF, quoting the clause.

When to use what

Approach	Use when	Effort
Prompting	General tasks; give context in the prompt	Minutes
RAG	Answers must come from your documents and stay current	Days
Fine-tuning	A fixed style/format or specialised task at scale	Weeks, needs labelled examples
Training from scratch	Almost never for a business	Huge

Agentic AI



An agent loops: plan → act with a tool → observe → check the goal → adjust, with human approval for risky actions.

Component	What it means
Goal	A task, not a single question: "Prepare the weekly sales review"
Planning	Breaks the goal into steps
Tools	Things it can use: SQL database, Python, Excel, web search, email, calendar, CRM
Memory	Keeps context across steps or sessions
Guardrails	Permissions, approvals, logging, spending limits

Chatbot vs copilot vs agent

Chatbot	Answers questions in a conversation
Copilot	Helps inside an app while you stay in control (Copilot in Excel, GitHub Copilot)
Agent	Takes multi-step actions on its own toward a goal (Excel Agent Mode, Claude Code, ChatGPT agent, Tableau Next agents)

Agentic examples for data work

- Data-cleaning agent: profiles a new file, fixes types, flags anomalies, writes a summary.
- Report agent: runs SQL every Monday, updates Power BI, writes commentary, emails stakeholders.
- Research agent: gathers market data from the web with citations into a spreadsheet.
- Coding agent: builds and tests a Python pipeline from a description.

> **Tip** **Model Context Protocol (MCP)** is an open standard for connecting AI assistants to tools and data sources (databases, Google Drive, GitHub, Slack) in a consistent way, so one agent can use many apps.

Using AI at work

High-value uses for analysts

Write and explain	Excel formulas, SQL, DAX, Python; explain someone else's code
Clean and classify	Standardise names, categorise free-text feedback, extract fields from PDFs
Analyse	Upload a CSV and ask for summaries, charts and tests (verify the numbers)
Communicate	Draft executive summaries, slides, emails in the right tone
Learn	Tutor mode: explain a concept, quiz you, review your answer

Risks and safe practice

✔ Verify facts and numbers; ask for sources; recompute key figures

✔ Never paste confidential, customer or personal data into unapproved tools (India's DPDP Act 2023 governs personal data)

✔ Use company-approved enterprise versions with data-protection terms

✔ Watch for bias in outputs about people

✔ Keep a human in the loop for decisions affecting customers or money

✔ Label AI-generated content where required

Glossary

Prompt · **system prompt** · **token** · **context window** · **embedding** · **vector database** · **RAG** · **fine-tuning** · **hallucination** · **temperature** · **multimodal** · **agent** · **tool use / function calling** · **MCP** · **guardrails** · **evaluation (evals)** · **open-weight model** (downloadable, e.g. Llama, Mistral) vs **closed model** (API only).

Interview Q&A

Q AI vs ML vs generative AI vs agentic AI?

AI is the broad field; ML learns from data; generative AI creates new content; agentic AI plans and acts with tools to complete goals.

Q How does an LLM generate text?

It splits text into tokens and repeatedly predicts the next token using a transformer trained on vast text, sampling one token at a time.

Q What is RAG and why do companies use it?

Retrieve relevant internal documents, then have the LLM answer from them. It reduces hallucinations, keeps answers current and cites sources without retraining.

Q What risks would you flag before using GenAI on customer data?

Privacy and consent (DPDP), data leaving the company, hallucinated outputs, bias, security (prompt injection), and lack of audit trails.

Q Give an example of an AI agent in analytics.

An agent that, every Monday, queries sales, refreshes the dashboard, spots anomalies, drafts a commentary and asks a human to approve before emailing.

Resources

- ▶ YouTube: Simplilearn generative AI full course · ▶ YouTube: Simplilearn agentic AI tutorial · ▶ YouTube: Simplilearn what is AI
- ▶ YouTube: Andrej Karpathy intro to large language models · ▶ YouTube: 3Blue1Brown large language models explained briefly
- ▶ YouTube: IBM Technology what is RAG · ▶ YouTube: IBM Technology what are AI agents
- 📘 Google: Introduction to Generative AI (free course) · 📘 DeepLearning.AI short courses
- 📘 Anthropic: Building effective agents · 📘 Model Context Protocol
- 📘 Microsoft: AI fundamentals learning path